

LAB 4

Solve below questions:

1. Method Overloading: Write a class Calculator with overloaded methods add(). Implement add() methods that take:

- Two integers
- Two double values
- Three integers
- A variable number of integers

PROGRAM:

```
package mDemo;
```

```
class Calculator {
```

```
    // Method to add two integers
```

```
    public int add(int a, int b) {  
        return a + b;  
    }
```

```
    // Method to add two double values
```

```
    public double add(double a, double b) {  
        return a + b;  
    }
```

```
    // Method to add three integers
```

```
    public int add(int a, int b, int c) {  
        return a + b + c;  
    }
```

```
    // Method to add a variable number of integers
```

```
    public int add(int... numbers) {  
        int sum = 0;  
        for (int num : numbers) {  
            sum += num;  
        }  
        return sum;  
    }
```

```
}
```

```
public class MethodOverloading {
```

```
    public static void main(String[] args) {  
        Calculator calculator = new Calculator();
```

```

// Test the methods
System.out.println("Sum of 2 and 3: " + calculator.add(2, 3));
System.out.println("Sum of 2.5 and 3.5: " + calculator.add(2.5, 3.5));
System.out.println("Sum of 1, 2, and 3: " + calculator.add(1, 2, 3));
System.out.println("Sum of 1, 2, 3, 4, and 5: " + calculator.add(1, 2, 3, 4, 5));
}
}

```

```

<terminated> MethodOverloading [Java Application] C:\Users\Ashi chouhan\.p2\pool\plugins\org.
Sum of 2 and 3: 5
Sum of 2.5 and 3.5: 6.0
Sum of 1, 2, and 3: 6
Sum of 1, 2, 3, 4, and 5: 15

```

2. Super Keyword: Create a class Person with a constructor that accepts and sets name and age.

- Create a subclass Student that adds a grade property and initializes name and age using the super keyword in its constructor.
- Demonstrate the creation of Student objects and the usage of super to call the parent class constructor.

PROGRAM:

```

package Lab4;

class Person {
    // Private variables for name and age
    private String name;
    private int age;

    // Parameterized Constructor that accepts name and age
    Person (String name, int age) {
        this.name = name;
        this.age = age;
    }

    public String getName() {
        return name;
    }

    public int getAge() {
        return age;
    }
}

class Student extends Person {

    private char grade;

    // Parameterized Constructor that accepts name and age using super keyword

```

```

        Student(String name, int age, char grade) {
            super(name,age);
            this.grade = grade;

        }

        public char getGrade() {
            return grade;
        }
    }

    public class SuperKeyword {

        public static void main(String[] args) {
            // Student class Object
            Student s1 = new Student("Ashi", 21, 'A');

            // Using Student class object to get name, age and grade
            System.out.println("The name of the student is :" + s1.getName() + "
And age is : " + s1.getAge());
            System.out.println("Grade : " + s1.getGrade());

        }

    }
}

```

OUTPUT:

```

The name of the student is :Ashi And age is : 21
Grade : A

```

3. Super Keyword: Create a base class Shape with a method draw() that prints "Drawing Shape".

- Create a subclass Circle that overrides draw() to print "Drawing Circle".
- Inside the draw() method of Circle, call the draw() method of the Shape class using super.draw().
- Write a main method to demonstrate calling draw() on a Circle object.

PROGRAM:

```

package Lab4;

class Shape {
    public void draw() {
        System.out.println("Drawing Shape");
    }
}

class Circle extends Shape {
    // Overriding draw() method of parent class Shape
    public void draw() {

```

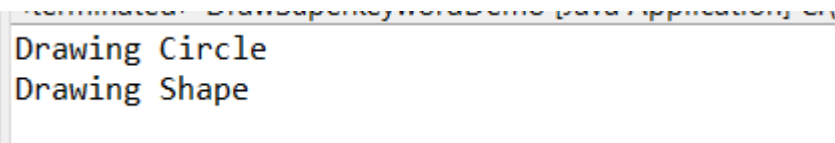
```

        System.out.println("Drawing Circle");
        // Calling Shape class draw() method using super keyword
        super.draw();
    }
}
public class DrawSuperkeywordDemo {

    public static void main(String[] args) {
        // Creating object of Circle class and calling draw() methos
        Circle c1 = new Circle();
        c1.draw();
    }
}

```

OUTPUT:



```

Drawing Circle
Drawing Shape

```

4. Write a Java Program to count the number of words in a String without using the Predefined method?

PROGRAM:

```

package Lab4;

public class WordCounter {

    // Function that returns the numbers of words in a string
    public static int wordCount(String str) {
        // To check is the string is empty or null
        if (str == null || str.isEmpty() ) {
            return 0;
        }

        // Initializing count
        int count = 0;
        // Initializing isSpace for a space character
        boolean isSpace = true;

        // Loop to iterate through the length of the string
        for ( int i = 0; i < str.length(); i++) {
            // if condition to check if the character at i index is not a
            space
            if(str.charAt(i) != ' ') {
                // Increment count if isSpace is true initialing and
                character is not a space
                if (isSpace == true) {
                    count++;
                }
                // set isSpace to false
                isSpace = false;
            }
        }
    }
}

```

```

        }
        else {
            // set isSpace to true if character at i is a space
character
            isSpace = true;
        }
    }

    return count;
}

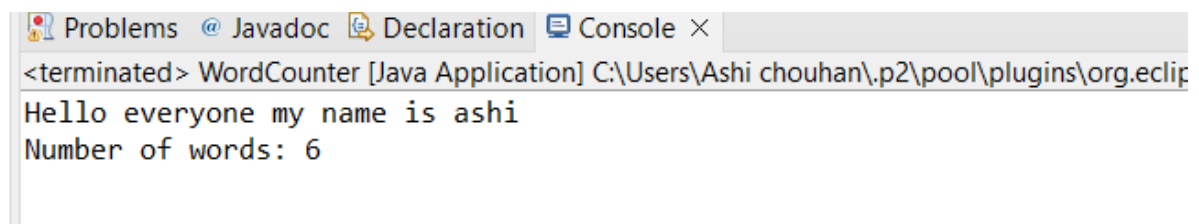
```

```

public static void main(String[] args) {
    String str = "Hello everyone my name is ashi";
    System.out.println(str);
    // Prints the Number of words
    System.out.println("Number of words: " + wordCount(str));
}
}

```

OUTPUT:



```

<terminated> WordCounter [Java Application] C:\Users\Ashi chouhan\p2\pool\plugins\org.eclipse
Hello everyone my name is ashi
Number of words: 6

```

5. Write a Java Program to remove all white spaces from a String?

PROGRAM:

```

package Lab4;

public class StringWhiteSpaceRemoval {

    public static StringBuilder removeSpace(String string1) {

        // if string is empty or null , return null
        if(string1.isEmpty() || string1 == null ) {
            return null;
        }

        // StringBuilder object to return the result as a string
        StringBuilder result = new StringBuilder();

        for(int i = 0; i<string1.length() ; i++) {
            // Append character at i index if character is not a white
space
            if(string1.charAt(i) != ' ') {
                result.append(string1.charAt(i));
            }
        }
    }
}

```

```

        return result;
    }

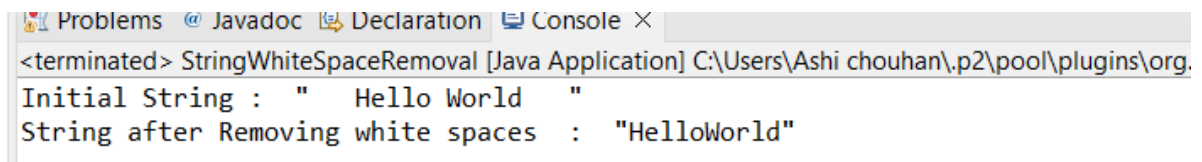
    public static void main(String[] args) {
        String string1 = "  Hello World  ";
        // Calling removeSpace function
        StringBuilder result = removeSpace(string1);

        System.out.println("Initial String : " + " \"" + string1 + "\"");

        System.out.println("String after Removing white spaces : " + " \"" +
+ result + "\"");
    }
}

```

OUTPUT:



```

<terminated> StringWhiteSpaceRemoval [Java Application] C:\Users\Ashi chouhan\p2\pool\plugins\org.
Initial String : "  Hello World  "
String after Removing white spaces : "HelloWorld"

```

6. WAP to find occurrence of given(substring) in the given string.

PROGRAM:

```

package Lab4;

public class SubStringOccurrenceCount {

    // Function to count occurrences of 'substring' in 'mainString'
    public static int countOccurrences(String mainString, String substring) {
        // return 0 if string or substring is empty or null
        if (mainString == null || substring == null || substring.isEmpty()) {
            return 0; // No occurrences if input is invalid
        }

        int count = 0;
        int index = 0;

        // Loop to find and count occurrences of the substring
        while ((index = mainString.indexOf(substring, index)) != -1) {
            count++;
            index += substring.length();
        }

        // Return the total count
        return count;
    }

    public static void main(String[] args) {
        // Declaring mainString and SubString
    }
}

```

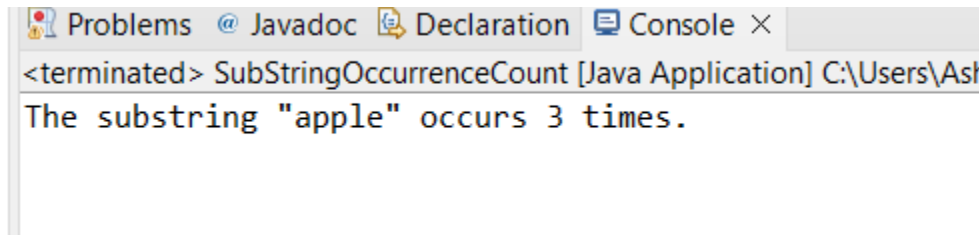
```

String mainString = "banana apple orange apple orange banana apple.";
String substring = "apple";

// Printing count of subString occurrences by calling countOccurrences
function
    int count = countOccurrences(mainString, substring);
    System.out.println("The substring \"" + substring + "\" occurs " + count +
" times.");
}
}

```

OUTPUT:



7. Write a java class to implement any 10 string methods:

- | | |
|---------------|--------------------|
| • Replace | • endsWith |
| • Contains | • EqualsIgnoreCase |
| • replaceAll | • toLowerCase |
| • indexOf | • toUpperCase |
| • substring | • isEmpty |
| • Equals | • Length |
| • lastIndexOf | • split |
| • startsWith | |

PROGRAM:

```

package Lab4;

import java.util.Arrays;

public class StringMethodsDemo {

    public static void main(String[] args) {
        String str = "Hello world";

        // replace()
        System.out.println("replace() : " + str.replace("w", "W"));

        // contains()
        System.out.println("contains() : " + str.contains("Hello"));

        // replaceAll()
        String str2 = str.replaceAll("world", "Ashi");
        System.out.println("replaceAll() : " + str2);

        // indexOf()
        System.out.println("indexOf() : " + str.indexOf("o"));

        // substring()
        System.out.println("substring() : " + str.substring(5));
    }
}

```

```

// equals()
System.out.println("equals() : " + str.equals(str2));

// lastIndexOf()
System.out.println("lastIndexOf() : " + str.lastIndexOf("o"));

// startsWith()
System.out.println("startsWith() : " + str.startsWith("Hello"));

// endsWith()
System.out.println("endsWith() : " + str.endsWith("Hello"));

// equalsIgnoreCase()
System.out.println("equalsIgnoreCase() : " + str.equalsIgnoreCase("HELLO
WORLD"));

// toLowerCase()
System.out.println("toLowerCase() : " + str.toLowerCase());

// toUpperCase()
System.out.println("toUpperCase() : " + str.toUpperCase());

// isEmpty()
System.out.println("isEmpty() : " + str.isEmpty());

// length()
System.out.println("length() : " + str.length());

// split()
String[] splitStr = str.split(" ");
System.out.println("split() : " + Arrays.toString(splitStr));
}
}

```

OUTPUT:

```

<terminated> StringMethodsDemo [Java Application] C:\Users\Ashi chouhan\
replace() : Hello World
contains() : true
replaceAll() : Hello Ashi
indexOf() : 4
substring() : world
equals() : false
lastIndexOf() : 7
startsWith() : true
endsWith() : false
equalsIgnoreCase() : true
toLowerCase() : hello world
toUpperCase() : HELLO WORLD
isEmpty() : false
length() : 11
split() : [Hello, world]

```

8. Write a java program to implement string tokenizer.

PROGRAM:

```
package Lab4;
import java.util.StringTokenizer;

public class TokenizerDemo {

    public static void main(String[] args) {

        String str = "Hello everyone , nice to meet you all.";

        // StringTokenizer object with comma as the delimiter
        StringTokenizer tokenizer = new StringTokenizer(str, ", ");

        // Printing all tokens
        System.out.println("Tokens:");

        // Loop to print all token is there are many tokens in a string
        while (tokenizer.hasMoreTokens()) {
            String token = tokenizer.nextToken();
            System.out.println(token);
        }
    }
}
```

OUTPUT:

```
<terminated> TokenizerDemo [Java Application] C:\Users\As
Tokens:
Hello
everyone
nice
to
meet
you
all.
```